CSE643 Monsoon Semester

Assignment 1: Truck Loading and Delivery Planning

Formulate and *implement* a program to plan truck-loading in a dispatch centre, modeling this using state-space search and compare the performance of different search algorithms.

The dispatch centre of a logistics company continuously receives packages and stacks them in the space available. These must must be loaded into one or more delivery trucks and dispatched for delivery. Each package has a destination represented by an integer (1, 2, 3, ...). Empty trucks arrive at the dispatch centre after completing their delivery runs and each has the same fixed capacity. At the start of the day all trucks are empty and available for loading.

Packages are unloaded from the front of a truck. Therefore, a package should ideally not be loaded in front of a package whose destination must be visited earlier. In other words, packages in each truck should respect the destination ordering as far as possible.

Your task is to **formulate** the problem and use state space search and/or the other algorithms discussed in class to implement a *truck-loading program* that **minimizes** a weighted sum of **(a) the total number of trucks** assigned to the dispatch centre, **(b) the average time to deliver a package**, i.e., the time takes from its arrival in the dispatch centre when it is unloaded at its destination, All destinations are equidistant from each other along the same highway leading away from the dispatch centre.

Generate random problems with varying numbers of packages, sequence of arrival, and destinations, and compare the performance of different algorithms (including the *heuristics* used) in terms of solution quality as well as search efficiency (nodes expanded).

Explore various approaches to modeling the problem as well as its variations as may be expected in real-life scenarios.

Summarize your recommended *approach* along with assumptions it relies on as well as your *experimental findings* into a crisp set of learnings that you can **recall** easily as well as **explain** when asked in the in-class assessment.